

```
In [26]: import numpy as np
```

```
In [27]: np.__version__
```

```
Out[27]: '1.26.4'
```

```
In [28]: import sys  
sys.version
```

```
Out[28]: '3.12.7 | packaged by Anaconda, Inc. | (main, Oct 4 2024, 13:17:27) [MSC v.192  
9 64 bit (AMD64)]'
```

```
In [29]: l=[12,52,67,8,3]  
l
```

```
Out[29]: [12, 52, 67, 8, 3]
```

```
In [30]: type(l)
```

```
Out[30]: list
```

```
In [31]: arr = np.array(l)  
arr
```

```
Out[31]: array([12, 52, 67, 8, 3])
```

```
In [32]: type(arr)
```

```
Out[32]: numpy.ndarray
```

```
In [33]: type(l)
```

```
Out[33]: list
```

```
In [34]: np.arange(10)
```

```
Out[34]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [35]: np.arange(20)
```

```
Out[35]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,  
17, 18, 19])
```

```
In [36]: np.arange(30)
```

```
Out[36]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,  
17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29])
```

```
In [37]: np.arange(10,20)
```

```
Out[37]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
In [38]: np.arange(-20, 19)
```

```
Out[38]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
               -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17, 18])
```

```
In [39]: np.arange(20,10) # first argument must be less then the second argument
```

```
Out[39]: array([], dtype=int32)
```

```
In [40]: np.arange(10,30,5) # 3rd argument is used as interval between values
```

```
Out[40]: array([10, 15, 20, 25])
```

```
In [41]: np.zeros(10)
```

```
Out[41]: array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0.])
```

```
In [42]: np.zeros(10, dtype=int) #hyperparameter tuning is when you change the datatype t
```

```
Out[42]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

```
In [43]: np.zeros((2,2), dtype=int)
```

```
Out[43]: array([[0, 0],
               [0, 0]])
```

```
In [44]: np.zeros((3,3), dtype=int)
```

```
Out[44]: array([[0, 0, 0],
               [0, 0, 0],
               [0, 0, 0]])
```

```
In [45]: zero = np.zeros([2,2])
print(zero)
print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
<class 'numpy.ndarray'>
```

```
In [46]: np.zeros((10,10), dtype=int)
```

```
Out[46]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [47]: np.ones((2,2))
```

```
Out[47]: array([[1., 1.],
               [1., 1.]])
```

```
In [48]: np.ones((2,2), dtype=int)
```

```
Out[48]: array([[1, 1],
               [1, 1]])
```

```
In [49]: rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[49], line 1
----> 1 rand(3,2)

NameError: name 'rand' is not defined
```

```
In [ ]: np.random.rand(3,2)
```

```
Out[ ]: array([[0.48807068, 0.9649831 ],
               [0.23487611, 0.3952803 ],
               [0.01156376, 0.48366172]])
```

```
In [ ]: np.random.rand(3,13)
```

```
Out[ ]: array([[0.13843751, 0.03995816, 0.86295295, 0.4078327 , 0.98447316,
                0.73688526, 0.47323894, 0.81814046, 0.11903866, 0.56661962,
                0.8978478 , 0.12550211, 0.65024215],
               [0.66344594, 0.68998414, 0.20957584, 0.50263513, 0.95280916,
                0.13456277, 0.66096799, 0.23184991, 0.60481125, 0.46587571,
                0.91742232, 0.33902058, 0.43971192],
               [0.25626807, 0.83318008, 0.47357606, 0.28262595, 0.56013129,
                0.96547912, 0.78981903, 0.74324144, 0.03121527, 0.81560506,
                0.95503715, 0.0680684 , 0.30109854]])
```

```
In [ ]: np.random.randint(3)
```

```
Out[ ]: 0
```

```
In [ ]: np.random.randint(1,10,2)
```

```
Out[ ]: array([5, 8])
```

```
In [ ]: np.random.randint(1,12,10)
```

```
Out[ ]: array([ 9,  7,  9,  9,  9, 11,  6,  5,  6,  6])
```

```
In [ ]: np.random.randint(10,40,((10,10)))
```

```
Out[ ]: array([[18, 19, 20, 21, 34, 37, 24, 28, 14, 17],
               [38, 21, 11, 37, 37, 33, 37, 26, 33, 28],
               [26, 20, 23, 13, 13, 17, 37, 14, 17, 14],
               [23, 21, 19, 25, 33, 33, 33, 17, 23, 22],
               [35, 33, 23, 36, 39, 35, 39, 14, 31, 19],
               [16, 14, 13, 37, 33, 15, 16, 24, 35, 22],
               [26, 18, 10, 14, 36, 20, 35, 14, 22, 21],
               [25, 15, 14, 14, 39, 17, 15, 35, 15, 13],
               [15, 36, 27, 13, 21, 34, 11, 22, 11, 31],
               [17, 26, 14, 39, 12, 34, 15, 16, 10, 14]])
```

```
In [ ]: np.arange(1,13)
```

```
Out[ ]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
```

```
In [ ]: np.arange(1,13).reshape(3,4) #reshape() is used to define the size of the matrix
```

```
Out[ ]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12]])
```

```
In [ ]: np.arange(1,13).reshape((4,3))
```

```
Out[ ]: array([[ 1,  2,  3],
               [ 4,  5,  6],
               [ 7,  8,  9],
               [10, 11, 12]])
```

```
In [ ]: np.arange(1,13).reshape(5,4)
#reshape arguments must be such that the product of reshape arguments must be eq
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[34], line 1
----> 1 np.arange(1,13).reshape(5,4)

ValueError: cannot reshape array of size 12 into shape (5,4)
```

slicing

```
In [ ]: b = np.random.randint(10,20,(4,5))
b
```

```
Out[ ]: array([[17, 19, 17, 17, 19],
               [10, 17, 10, 12, 11],
               [12, 16, 18, 15, 16],
               [15, 13, 14, 16, 15]])
```

```
In [ ]: type(b)
```

```
Out[ ]: numpy.ndarray
```

```
In [ ]: b
```

```
Out[ ]: array([[17, 19, 17, 17, 19],
               [10, 17, 10, 12, 11],
               [12, 16, 18, 15, 16],
               [15, 13, 14, 16, 15]])
```

```
In [ ]: b[:]
```

```
Out[ ]: array([[17, 19, 17, 17, 19],
               [10, 17, 10, 12, 11],
               [12, 16, 18, 15, 16],
               [15, 13, 14, 16, 15]])
```

```
In [ ]: b[1:3] #will print rows only
```

```
Out[ ]: array([[10, 17, 10, 12, 11],
               [12, 16, 18, 15, 16]])
```

```
In [ ]: b[1,3] # will give one value from the matrix
```

Out[]: 12

In []: `b[1,-1]`

Out[]: 11

In []: `b[2:3]`

Out[]: `array([[12, 16, 18, 15, 16]])`

In []: `b[0:-2]`

Out[]: `array([[17, 19, 17, 17, 19],
[10, 17, 10, 12, 11]])`

In []: `b`

Out[]: `array([[17, 19, 17, 17, 19],
[10, 17, 10, 12, 11],
[12, 16, 18, 15, 16],
[15, 13, 14, 16, 15]])`

operations

In []: `a2 = np.random.randint(0,100,(10,10))`
`a2`

Out[]: `array([[52, 90, 91, 67, 86, 49, 19, 1, 74, 11],
[8, 0, 28, 28, 80, 77, 99, 79, 79, 59],
[54, 39, 98, 7, 75, 80, 23, 61, 50, 19],
[18, 7, 48, 56, 11, 75, 55, 48, 83, 83],
[75, 24, 50, 80, 47, 96, 21, 5, 58, 79],
[85, 21, 35, 48, 49, 58, 79, 57, 87, 27],
[51, 14, 73, 56, 81, 74, 17, 17, 79, 21],
[46, 40, 2, 13, 79, 20, 1, 73, 84, 32],
[31, 84, 33, 53, 87, 1, 54, 96, 96, 60],
[82, 79, 98, 94, 47, 7, 25, 58, 87, 91]])`

In []: `a2[::-2]`

Out[]: `array([[82, 79, 98, 94, 47, 7, 25, 58, 87, 91],
[46, 40, 2, 13, 79, 20, 1, 73, 84, 32],
[85, 21, 35, 48, 49, 58, 79, 57, 87, 27],
[18, 7, 48, 56, 11, 75, 55, 48, 83, 83],
[8, 0, 28, 28, 80, 77, 99, 79, 79, 59]])`

In []: `a2[::-3]`

Out[]: `array([[82, 79, 98, 94, 47, 7, 25, 58, 87, 91],
[51, 14, 73, 56, 81, 74, 17, 17, 79, 21],
[18, 7, 48, 56, 11, 75, 55, 48, 83, 83],
[52, 90, 91, 67, 86, 49, 19, 1, 74, 11]])`

In []: `a2[0:10:3]`

```
Out[ ]: array([[52, 90, 91, 67, 86, 49, 19, 1, 74, 11],
               [18, 7, 48, 56, 11, 75, 55, 48, 83, 83],
               [51, 14, 73, 56, 81, 74, 17, 17, 79, 21],
               [82, 79, 98, 94, 47, 7, 25, 58, 87, 91]])
```

```
In [ ]: a2.max()
```

```
Out[ ]: 99
```

```
In [ ]: a2.min()
```

```
Out[ ]: 0
```

```
In [ ]: a2.mean()
```

```
Out[ ]: 52.84
```

indexing

```
In [51]: mat = np.arange(0,100).reshape(10,10)
         mat
```

```
Out[51]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [52]: row = 4
         col = 5
         col
```

```
Out[52]: 5
```

```
In [53]: row
```

```
Out[53]: 4
```

```
In [54]: mat
```

```
Out[54]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [55]: mat[row,col] # will give the value at the specified row and column
```

```
Out[55]: 45
```

```
In [56]: mat[4,5]
```

```
Out[56]: 45
```

```
In [57]: mat[:,]
```

```
Out[57]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [58]: mat[:,5] # will give all the values in the 5th column
```

```
Out[58]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
In [60]: mat[row,:]
```

```
Out[60]: array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [61]: mat[:,col]
```

```
Out[61]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
In [62]: mat[:row]
```

```
Out[62]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [63]: mat[row:]
```

```
Out[63]: array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [64]: mat[:,8]
```

```
Out[64]: array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [65]: mat[:, -1]
```

```
Out[65]: array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [66]: mat
```

```
Out[66]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [ ]: mat[3:-3] # will give all the rows except the first 3 and last 3 rows
```

```
Out[ ]: array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [68]: mat[0]
```

```
Out[68]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [69]: mat[5:7]
```

```
Out[69]: array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [70]: mat[0:10:3]
```

```
Out[70]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [71]: mat[:, :-1]
```

```
Out[71]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
                [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [72]: mat[:, :-2]
```

```
Out[72]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
                [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
                [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
                [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
                [10, 11, 12, 13, 14, 15, 16, 17, 18, 19]])
```



```
In [73]: mat[2:6]
```

```
Out[73]: array([[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [74]: mat
```

```
Out[74]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [76]: mat[2:6,2:5] # will give the values from 2nd to 5th row and 2nd to 5th column
```

```
Out[76]: array([[22, 23, 24],
               [32, 33, 34],
               [42, 43, 44],
               [52, 53, 54]])
```

```
In [77]: mat[1:4, 2:5] # will give the values from 1st to 4th row and 2nd to 5th column
```

```
Out[77]: array([[12, 13, 14],
               [22, 23, 24],
               [32, 33, 34]])
```

```
In [78]: mat[3:8, 2:5] # will give the values from row 3 to 7 and column 2 to 4
```

```
Out[78]: array([[32, 33, 34],
               [42, 43, 44],
               [52, 53, 54],
               [62, 63, 64],
               [72, 73, 74]])
```

Masking

```
In [79]: mat
```

```
Out[79]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [80]: id(mat)
```

Out[80]: 2651541344336

In [81]: `mat<50 # will give a boolean matrix where the values are True if the value is le`

Out[81]: array([[True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False]])

In [82]: `mat>50 # will give a boolean matrix where the values are True if the value is gr`

Out[82]: array([[False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, False, False, False, False, False, False, False, False, False,
 False],
 [False, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True],
 [True, True, True, True, True, True, True, True, True, True,
 True]])

In [83]: `mat == 50 # will give a boolean matrix where the values are True if the value is`

```
Out[83]: array([[False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [ True, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False],
                [False, False, False, False, False, False, False, False, False,
                False]])
```

```
In [84]: mat[mat == 50] # will give the values that are equal to 50
```

```
Out[84]: array([50])
```

```
In [86]: x = mat[mat < 50] # will give the values that are less than 50
x
```

```
Out[86]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
                34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [87]: y = mat[mat > 50] # will give the values that are greater than 50
y
```

```
Out[87]: array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
                68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
                85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

```
In [88]: a = mat[mat <= 50] # will give the values that are less than or equal to 50
a
```

```
Out[88]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
                34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

```
In [ ]:
```